# App Dev Project Report

## 1. Student Details

**Name:** Bhaven Sandeep Gupta
**Roll Number:** 24f3000936
**Email:** 24f3000936@ds.study.iitm.ac.in
**About Me:** I am a Student at IIT Madras BS Degree program with a deep interest in web application development, data-driven technologies and AI related applications. I enjoy building meaningful applications that combine learning, analytics, and user experience.

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## 2. Project Details

### Project Title: Hospital Management System

### Problem Statement:

Hospital Management System (HMS) web application that allows Admins, Doctors, and Patients to interact with the system based on their roles.

### Approach:

The app was built using Flask as the backend framework with a modular structure. It allows patients to book an appointment for the registered doctors in the respective department with ease. The admin is pre-defined in this app who can manage the doctors and patients and also keep track of the upcoming and past appointments.

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## 3. AI/LLM Declaration

No use of AI/LLM.

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# 4. Technologies and Frameworks Used

| Technology / Library | Purpose |
|---|---|
| **Flask** | Core backend web framework |
| **SQLAlchemy** | Object Relational Mapper for SQLite database |
| **Jinja2** | Template engine for rendering dynamic HTML pages |
| **Bootstrap 5** | Frontend styling and responsive design |
| **SQLite** | Lightweight local database for storing user data |

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# 5. Database Schema / ER Diagram

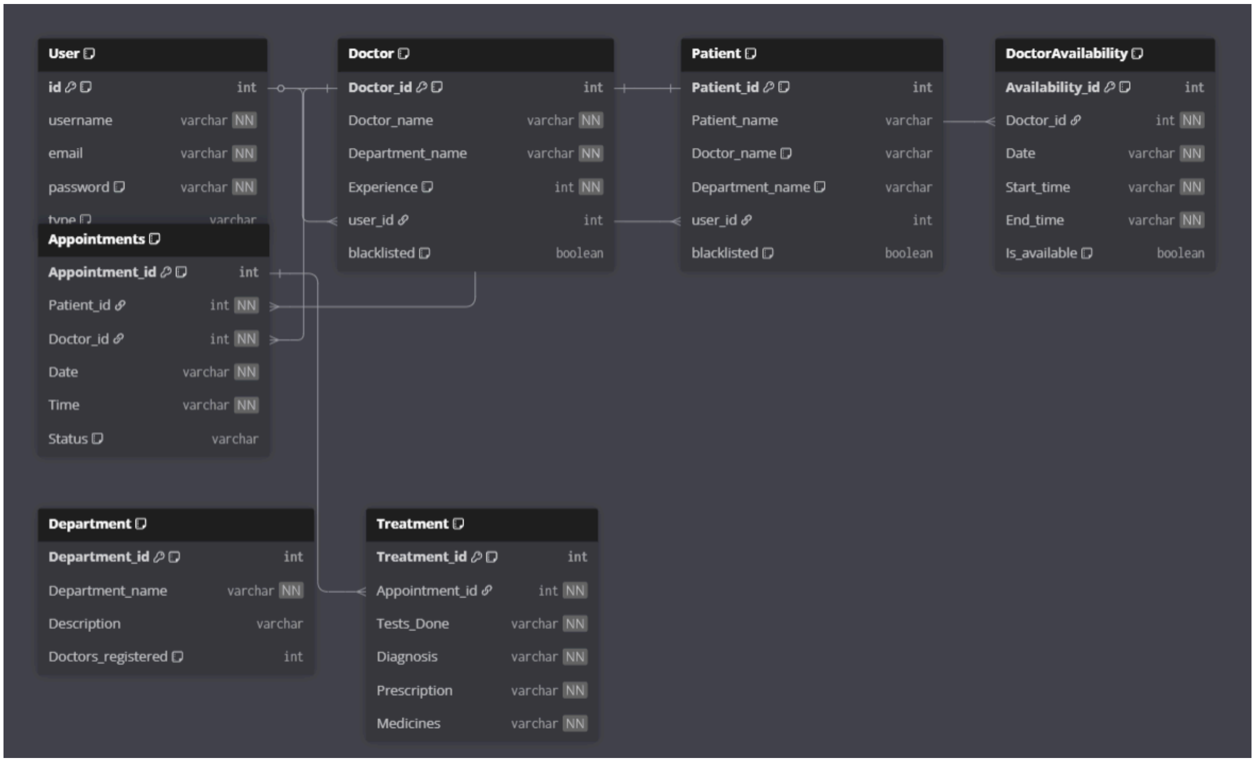
**Tables:**

1. **User** — Stores user profile details (id, username, email, password, type)
2. **Appointments** — Stores the appointments of the patients and doctors (Appointment_id, Patient_id, Doctor_id, Date, Time, Status)
3. **Treatment** — Stores the treatment history of a patient after an appointment (Treatment_id, Appointment_id, Tests_Done, Diagnosis, Prescription, Medicines)
4. **Department** — Stores the department names and doctors registered in that department (Department_id, Department_name, Description, Doctors_registered)
5. **Doctor** — Stores the information of a doctor (Doctor_id, Doctor_name, Department_name, Experience, blacklisted)
6. **Patient** — Stores the information of a patient (Patient_id, Patient_name, Doctor_name, Department_name, blacklisted)
7. **DoctorAvailability** — Stores the appointment details of a doctor (Availability_id, Doctor_id, Date, Start_time, End_time, Is_available)

**Relationships:**

- One-to-many: Doctor – DoctorAvailability
- One-to-many: Doctor – Appointments
- One-to-many: Patient – Appointments

ER Diagram:

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# 8. Video Presentation

**Drive Link:**

**App Dev 1 Project.mp4**

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